

Argos — Visual Speech Recognition

Built-In Confidence. Ready to Deploy.

Yoad Oxman · Argos / The Orchard · April 2026



We build production-grade visual speech recognition.

Lip-reading at scale, with built-in confidence.

WHAT WE DO

End-to-end pipeline: face detection, mouth cropping, visual feature extraction, language decoding — and confidence scoring on every word.

WHO USES IT

Teams that need to recover speech from footage where the audio is missing, noisy, or untrustworthy: media, broadcast, accessibility, investigations.

WHY IT MATTERS

Most speech-to-text systems hand you a transcript and a hope. We hand you a transcript with a per-word trust signal so you know what to review.



What we built — concretely

Six things actually exist today, end-to-end, on real data.

1. PIPELINE

Nine automatic stages from raw video to confidence-scored transcript. Containerized; runs anywhere with a GPU.

2. WEB UI

Drag-and-drop, live progress, downloadable HTML report. No command line required.

3. MODEL

Visual-only encoder + LLaMA-2 decoder, fine-tuned on lip-reading. Same architecture VSP-LLM published, our own training run.

4. CONFIDENCE

Per-word green/yellow/red. Per-segment Intelligibility Score. Hallucination auto-flagging.

5. EVALUATION

1,497-segment baseline with WER, IS, and expert-judge agreement metrics. Reproducible from raw inputs.

6. INTEGRATION

On-premise install, cloud deployment, or both. We deploy and hand off.

Everything you'll see today is in production today — not a 'with more research it could...' projection.



What is visual speech recognition?

Reading speech from video alone — no audio.

Same problem a deaf lip reader solves, automated and at scale.

WHEN AUDIO ISN'T USABLE

MUTED

Surveillance footage, archive video, clips with no recorded audio.

NOISY

Crowds, machinery, wind, overlapping speakers.

UNRELIABLE

Suspected dub, voice-over, mismatched lip sync.

AUDIO-LESS

Silent film, distant CCTV, streams that lost audio.

We've shipped this end-to-end. The next slides show the numbers.



Three numbers, in plain English

62%

useful output

Six of every ten segments deliver usable text
— viewers can extract meaning even when
the wording isn't perfect.

23%

clearly conveyed

About one in four segments needs no
review — delivered cleanly with no further
work.

1 in 5

auto-flagged

The system flags low-confidence segments
before you ever see them. You review the
flagged ones, not all of them.

Measured on 1,497 segments of unfiltered real-world video — varied lighting, head angles, multi-speaker scenes, no curation.

Validated against an independent expert reviewer.



What you'll see today

- Live demo — the system processing a real video, end-to-end
- Output gallery — what good, partial, and flagged segments look like
- How confidence works — per-word color coding and per-segment scoring
- How we validated it — independent expert agreement
- What's next — pre-processing, stronger model, Arabic, integration



Watch the system process a real video

Nothing here is pre-cached. The pipeline runs in your browser. Drag in, walk away, come back to color-coded results.

Press play on the next slide to start the walkthrough.



Live UI walkthrough

End-to-end demo recorded by the presenter.

Approximately 4-5 minutes · press play in PowerPoint to start

If video does not embed before the meeting, narrate the demo: drag-drop upload → 9-stage pipeline → color-coded report.



What you just saw

- Drag-drop upload — no command line, no scripting
- Nine automatic stages — face detection, mouth cropping, recognition, decoding
- Per-word color coding — green confident, yellow review, red likely error
- Per-segment Intelligibility Score — calibrated against expert judgment
- Output is a downloadable HTML report you can hand a reviewer



OUTPUT EXAMPLES

What good, partial, and flagged segments look like



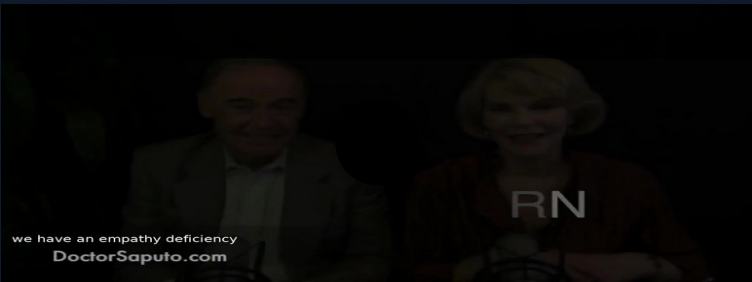
Real outputs — pick any tile to play

Six real segments decoded by the model. Click a tile in PowerPoint to play.



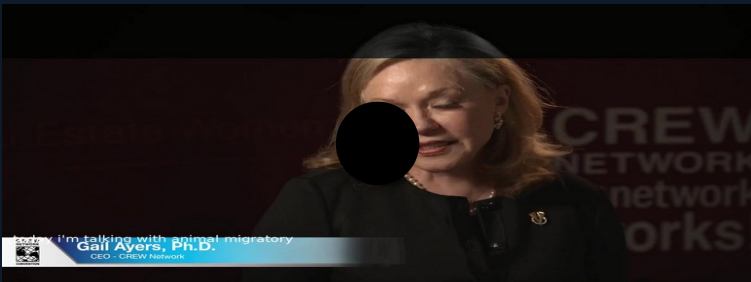
PERFECT

Clean visual, clean output



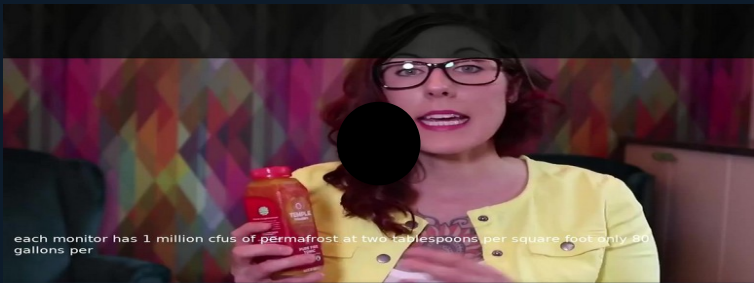
ALMOST PERFECT

Near-verbatim transcription



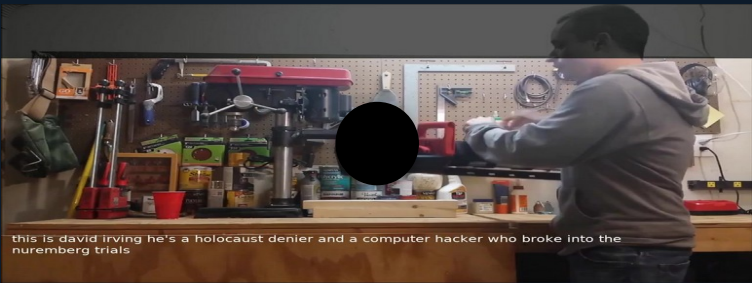
PARTIAL

Right gist, wrong specifics



NEAR MISS

Off by a phrase, recoverable with context



HALLUCINATION

Fluent but wrong — auto-flagged



TOPIC DRIFT

Model lost the thread, system flags it

These six are a deliberate spread — best-case to worst-case. All visually decoded, no audio, on real-world video.



What good, partial, and flagged look like

Three real segments from the Obama bin Laden announcement, decoded by the live model, color-coded by real per-word confidence.

CLEAN

Segment 14

WER 0%. All confident.

PARTIAL

Segment 31

WER 22%. Substitutions in yellow.

FLAGGED

Segment 5

WER 45%. Hallucination caught.



Example 1 — Clean speech, perfect transcription

Obama bin Laden announcement · segment #14 · 41.95–45.55 s

REFERENCE

and our allies over the last 10 years thanks to the tireless and heroic work of our military and our counterterrorism professionals we've made great strides in that effort

HYPOTHESIS

and our allies over the last 10 years thanks to the tireless and heroic work of our military and our counterterrorism professionals we've made great strides in that effort

WER 0%. 27 of 29 words at high confidence. No reviewer intervention needed.

GREEN: confident YELLOW: review RED: likely error



Example 2 — Partial recovery, model knows where it slipped

Obama bin Laden announcement · segment #31 · 92.90–96.50 s

REFERENCE

as president bush did shortly after 9 11 that our war is not against islam bin laden was not a muslim leader he was a mass murderer of muslims indeed al qaeda has slaughtered so many muslims

HYPOTHESIS

president bush said shortly after 9 11 that our war is not against islam bin laden was not a muslim leader he was a mass murderer of muslims then al qaeda slithered

WER 22%. "president bush did" became "president bush said". Substitutions appear in yellow — reviewer goes straight to them.

GREEN: confident YELLOW: review RED: likely error



Example 3 — Flagged before it ever reached you

Obama bin Laden announcement · segment #5 · 14.98–18.58 s

REFERENCE

heroic citizens saved even more heartbreak and destruction and yet we know that the worst images are those that were unseen to the world the empty seat at the dinner table

HYPOTHESIS

rwanda's genocide even more heartbreaking is russia and yet we know that the worst images are those that were unseen in the world mt killer whale

WER 45%. The model fabricated "rwanda's genocide" — and knew it. Lowest-confidence word at $p=0.02$. The system flags the line before the reviewer ever sees the transcript.

GREEN: confident YELLOW: review RED: likely error



CONFIDENCE & TRUST

How the system tells you what to trust



How do you know when to trust an output?

Most speech-to-text systems hand you a transcript. You read all of it. If something's wrong, you find out the hard way.

Argos is different. Every word and every segment carries a confidence signal. You read what's flagged — not everything.



Two layers of confidence

1. PER-WORD

From the model itself

- Every predicted word has a probability the model assigned it
- We surface those as green / yellow / red inline in the report
- You see exactly where the model was unsure

2. PER-SEGMENT

From the score system

- Intelligibility Score — a 0–5 score per segment
- Predicts whether a viewer will understand the output
- Calibrated against an independent expert reviewer



Per-word color coding on a real example

Argos VSP — Visual Speech Recognition

Client demo · Obama bin Laden announcement (May 1, 2011) · 6 segments

73%
OVERALL CONFIDENCE

SEGMENTS
6

SOURCE
Public TV broadcast

AUDIO
Not used — vision only

Per-word confidence: high review likely error · Hover any word to see its confidence value.

Obama bin Laden speech_2m30s_1min · segment #4 · 12.0s-15.6s

REF

office i directed leon panetta the director of the cia to make the killing or capture of bin laden the top priority of our war against al qaeda even as we continued our broader

HYP

office i directed the head of the cia to make the killing or capture of bin laden the top priority of our war against al qa

SYNTHETIC CONFIDENCE (B3 DATA PENDING)

Obama bin Laden speech_2m30s_1min · segment #0 · 0.0s-3.6s

REF

claudes wrong

HYP

made great strides in that effort we've disrupted terrorist attacks and strengthened our homeland defense in afghanistan we removed the

SYNTHETIC CONFIDENCE (B3 DATA PENDING)

Obama bin Laden speech_2m30s_1min · segment #1 · 3.0s-6.6s

REF

bin laden and al qaeda safe haven and support and around the globe we worked with our friends and allies to capture or kill scores of al qaeda terrorists including several who were a part of the 9 0 1 war

HYP

laden and al qaeda safe havens had support and around the globe we worked with our friends and allies to capture or kill

SYNTHETIC CONFIDENCE (B3 DATA PENDING)

Obama bin Laden speech_2m30s_1min · segment #2 · 6.0s-9.6s

REF

several who were a part of the 9 11 plot yet osama bin laden avoided capture and escaped across the afghan border into pakistan meanwhile al qaeda's forces were in the air

HYP

who were a part of the 9 11 plot yet osama bin laden evaded capture and escaped across the afghan border into

SYNTHETIC CONFIDENCE (B3 DATA PENDING)

Obama bin Laden speech_2m30s_1min · segment #3 · 9.0s-12.6s

REF

meanwhile al qaeda continued to operate from along that border and operate through its affiliates across the world and so shortly after taking office i directed the al qaeda's efforts to

HYP

meanwhile al qaeda continued to operate from along that border and operate through its affiliates across the world so shortly after taking office i directed

SYNTHETIC CONFIDENCE (B3 DATA PENDING)

Per-word confidence —

GREEN: confident

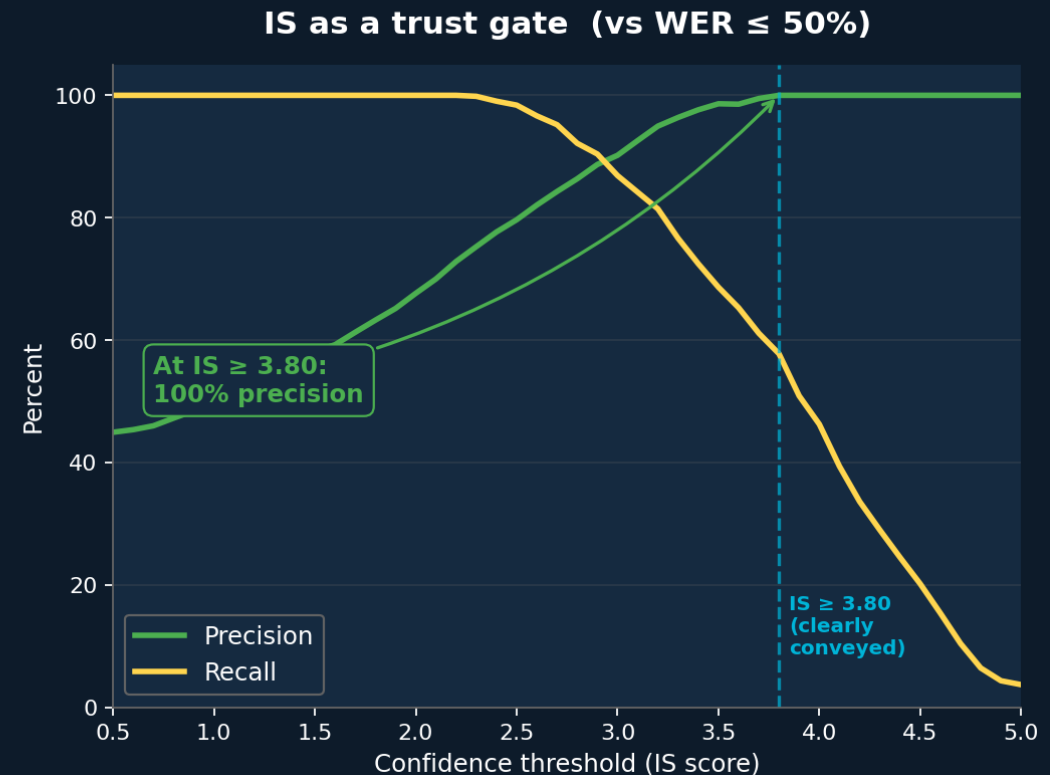
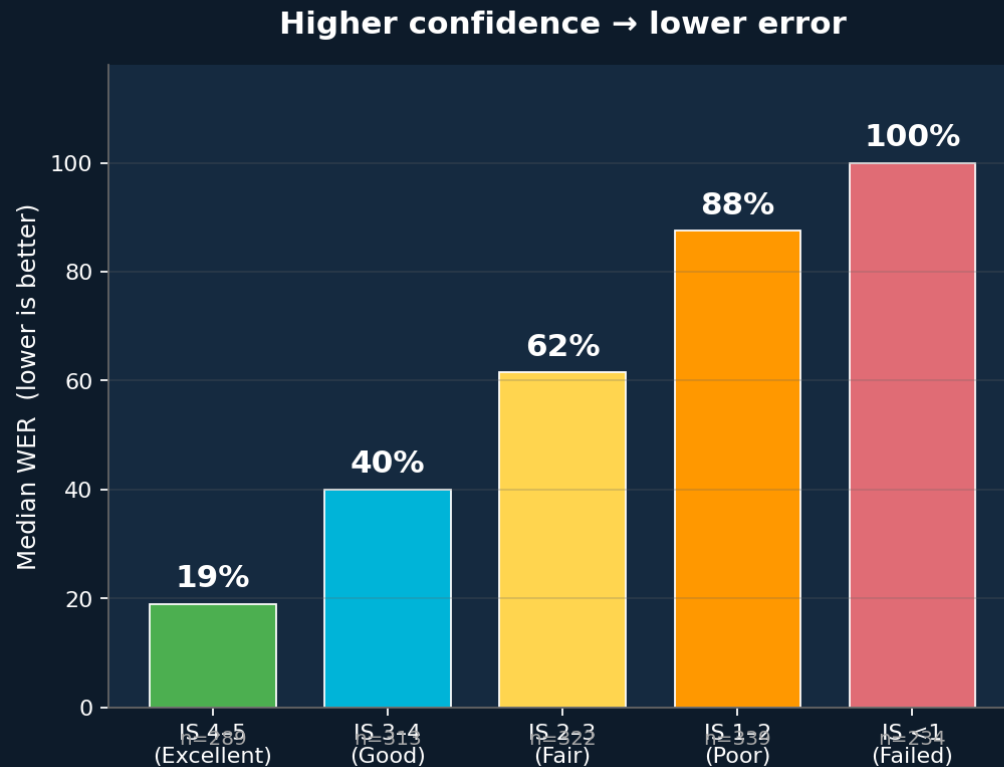
YELLOW: review

RED: likely error



Higher confidence → lower error, every time

Tier-1 confidence is real — n=1,497 real-world segments



When the system rates a segment "clearly conveyed," it really is — 100% precision across 1,497 real-world segments.



62% useful — on real-world video, not benchmark data

62%

of segments deliver useful output today

Measured on 1,497 segments of unfiltered real-world video

WHAT THAT 62% IS UP AGAINST

- Real YouTube footage — not a curated benchmark dataset
- Varied lighting, head angles, and camera distances
- Multi-speaker scenes, off-axis faces, partial occlusion
- Domain vocabulary the model has never seen
- No pre-filtering — the system processes whatever was uploaded

On segments the system flags as high-confidence, performance lifts further — we'll measure that conditional rate in the next iteration.



When the model fabricates, the system flags it

1 in 5

segments auto-flagged

before they ever reach you

- When the model produces fluent text it isn't sure about, we catch it
- Detection rules combine length anomalies and per-token confidence
- These are the dangerous failures — fluent but wrong
- You don't have to find them. The system finds them for you.

Detection combines length-anomaly rules with per-token confidence — the model can fabricate fluent text but it usually 'knows' it.



What this means for your workflow

- You review the flagged segments — not all of them. Hours saved per video.
- Reviewers go straight to the suspicious words instead of skimming text.
- Confidence is auditable — every claim has a number behind it.
- Bad output is a known unknown, not a hidden surprise.



VALIDATION

An independent expert agreed in 8 of 10 cases



How we validated the trust signals

- 1,497 real-world segments — not benchmark data
- Independent expert reviewer scored every one
- We compared the system's confidence signals to that judgment
- Result: the trust score and the expert agree on most segments

Real-world YouTube footage — varied lighting, head angles, multi-speaker scenes, accents. Not curated lab footage.



Agrees with the expert reviewer 82% of the time



Validated on the same 1,497 segments — "any useful meaning" judgment.



The trust signal stays stable

- We changed decode settings 16 different ways
- The trust signal moved less than a percentage point
- Translation: it isn't a fluke of one specific run

What this means: the confidence numbers you see in the demo are the same confidence numbers you'd see on your data, on your infrastructure, on your settings.

Tested across 16 different decode configurations on the same 1,497 segments.



An independent expert agreed 82% of the time

Not validated by us. Validated by an independent reviewer that didn't see our code, our scores, or our reasoning. That's the credibility behind every confidence number on every slide.



PIPELINE & ARCHITECTURE

What runs under the hood — light touch



How It Works: Data Flow

1

Video Frames — 25 fps raw video input

2

Mouth Crop — 96×96 pixel region around lips

3

Visual Features — AV-HuBERT encoder → 1024-dim vectors

4

Projection — Linear layer: 1024 → 4096-dim (LLM input space)

5

LLM Generates Text — LLaMA-2-7B decodes features into words

Visual encoder is frozen (pre-trained on LRS3). Only projection + LoRA adapters are trained.



Why lip reading is hard

Many words look identical on the lips. The model has to choose.

LOOK IDENTICAL

pat • bat • mat

Plosive sounds — same lip closure

LOOK IDENTICAL

thin • fin • shin

Fricatives — invisible airflow

LOOK IDENTICAL

see • zee • thee

Voicing — invisible vibration

The system uses language context and confidence scoring to pick the right one — and tells you when it can't.



Three components, end-to-end

From a raw video frame to a confidence-scored transcript.

1.

FACE DETECTION & MOUTH CROPPING

Find the speaker. Crop a stable, centered window around the mouth across every frame.

2.

VISUAL FEATURE EXTRACTION

Convert the mouth movement into a compact representation the language model can read.

3.

LANGUAGE DECODING

Generate the transcript word-by-word. Capture per-word confidence as we go.



Deploys where you need it

Same pipeline, two deployment paths. You pick.

CLOUD

Hosted on your AWS / GCP / Azure account. We run the GPU infra; you pay per minute of video.

- Faster setup, no hardware procurement
- Scales elastically with your queue
- We patch and update the stack

ON-PREMISE

Containerized install on your hardware. Your data never leaves your network.

- Air-gapped operation supported
- GPU on your own machines (T4 minimum)
- We support and maintain remotely



Live Demo

We will now run the full pipeline live

Upload a video → 8-stage processing → Per-segment results with IS scores

Drag & Drop

No command line

8 Stages

Fully automatic

IS Scoring

Per-segment quality

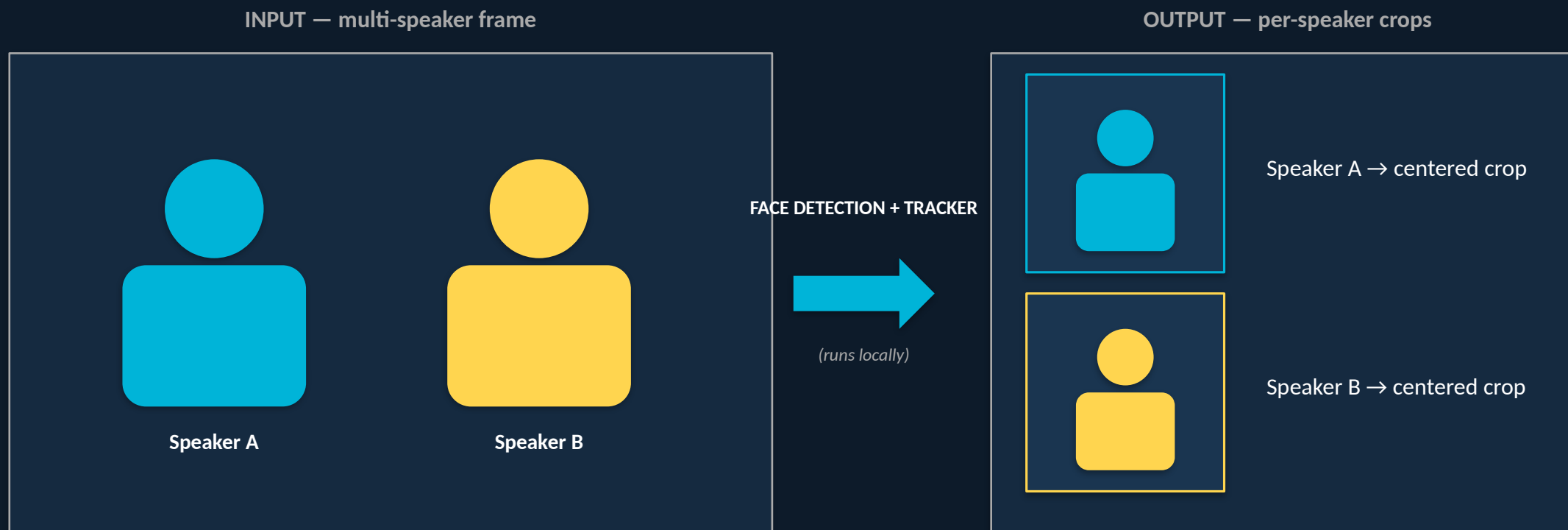


PRE-PROCESSING ROADMAP

Making it better starts before the model runs



Pre-processing 1 — Per-speaker entity split



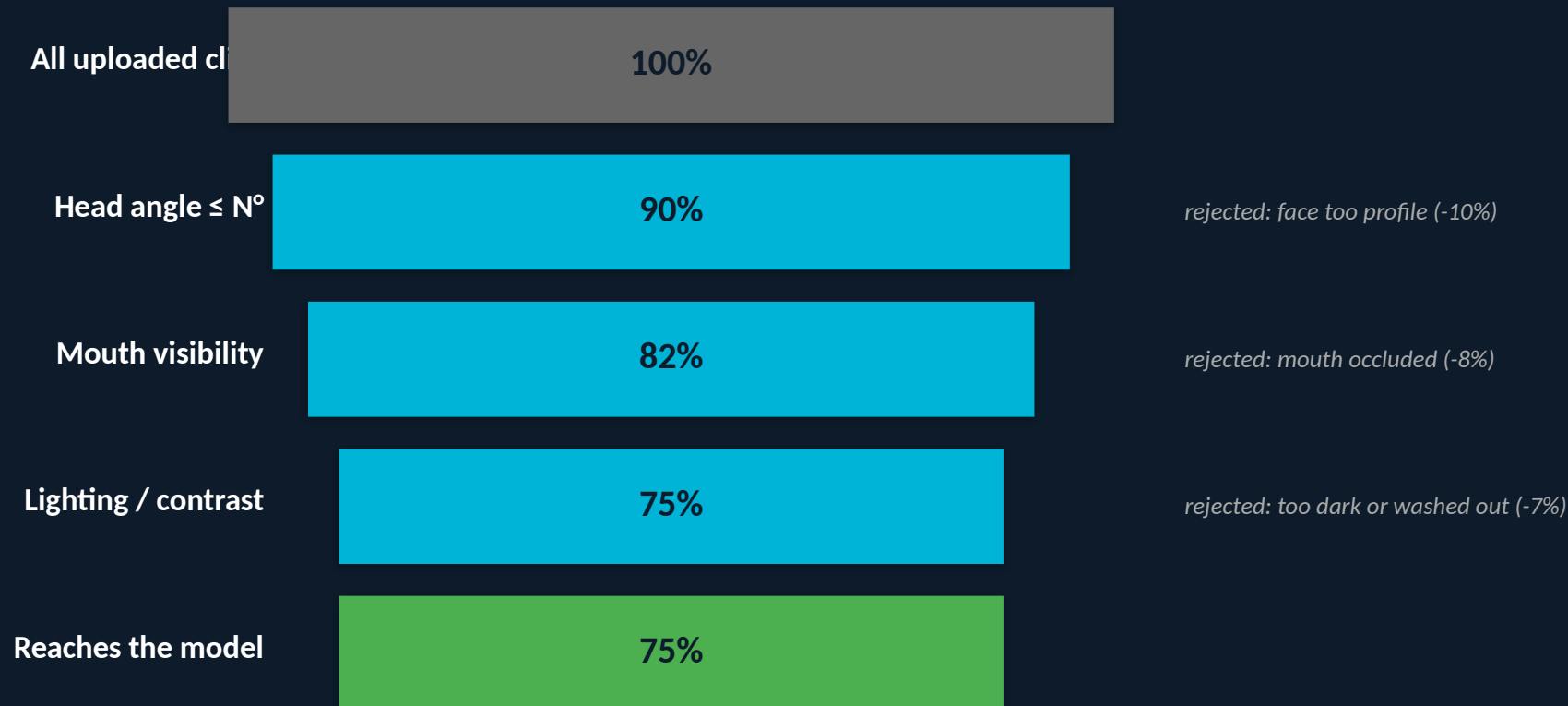
Two speakers in one frame become two centered crops. Each speaker gets its own clean lip-reading pass.

Status: planned ablation. Runs locally. No cloud dependency.



Pre-processing 2 — Quality pre-filter

Reject bad clips before they reach the model.



Three frame-level CV checks, all running locally. Status: planned ablation. Percentages illustrative — actual rejection rates depend on your video conditions.



Why pre-processing is the highest-leverage work

The model is never tested in isolation. What reaches it is what matters.

- Both stages run on your local machine — no extra cloud cost
- Both reduce wasted compute on unusable footage
- Both ship as clear ablation studies with measurable impact
- Together: a more reliable, more efficient pipeline by default



ROADMAP & INVESTMENT

Where we're going and what it takes to get there



Where we are vs. where we're going

TODAY

Working pipeline, validated trust signals, deployable today.

NEXT

Stronger model, more training data, pre-processing live.

AFTER

Domain-specialized fine-tunes for your content.



A stronger model, same architecture

The same pipeline. A larger, better-trained brain on the back.

- Drop-in upgrade of the language model that produces the final transcript
- Better grasp of context, named entities, and uncommon vocabulary
- No change to your integration — same inputs, same outputs, better text
- Requires retraining — not a switch, an investment

Biggest expected lift: yellow / red segments — where richer language priors help disambiguate visemically-identical words.



Today's data is a slice. There's a ceiling.

- We've identified the data ceiling on the current dataset
- Expanding the training set is the next unlock
- More data → better domain coverage → fewer flagged segments
- Especially powerful when paired with the stronger model

Direction matters more than the exact number here — we know which way the curve points.



Arabic Pipeline: Replication Roadmap

What's Needed & How We'll Do It

- **Arabic AV-HuBERT encoder (BOTTLENECK)**
- Arabic has different phonemes (pharyngeals, emphatics) — without fine-tuning, Arabic lips map to wrong clusters
- **Arabic LLM backend**
- Swap Llama-2 for Arabic-capable LLM (e.g. Jais, AceGPT, or Arabic-tuned Llama 3)
- **Arabic evaluation dataset (UNKNOWN)**
- No Arabic lip-reading benchmark exists — needs native speakers for MSA + dialect coverage
- **Training infrastructure**
- AWS GPU instance (existing) for AV-HuBERT fine-tuning and K-means reclustering
- **RTL text & normalization**
- RTL handling, spaCy Arabic, diacritic handling, Arabic NER — maturity

Practical Timeline

Step	Effort	Risks / Unknowns
AV-HuBERT fine-tune + K-means	5–10 weeks	Arabic visual data quality unknown
Arabic LLM swap	1–2 weeks	Tokenizer quality varies by model
Eval dataset	4–8 weeks	No benchmark exists; needs native speakers
RTL normalization + testing	3–6 weeks	RTL handling + end-to-end validation

Realistic estimate: 2–3 months (encoder pre-training is the bottleneck)

Pipeline code is language-agnostic. Main bottlenecks: encoder pre-training data and eval dataset collection. No Arabic lip-reading benchmark exists.



What it takes to ship the next model

To train the next model on the data scale needed, we need a training-budget commitment.

Compute

GPU hours for full training, not a quick partial pass.

Data

Curated and labeled domain segments — your domain.

Engineering

Pipeline integration, testing, and on-prem handoff.

Specific numbers in the follow-up — today is direction, not invoice.



INTEGRATION & NEXT STEPS

How we deploy this on your infrastructure



Three things to take with you

IT WORKS

Real pipeline, finished UI, usable today.

YOU CAN TRUST IT

Two layers of confidence — per word and per segment.

WE DEPLOY IT

End-to-end on your infrastructure, not an API call.



We deploy this on your infrastructure

You're not buying an API call. You're buying a working pipeline that runs in your environment.

1. INSTALL

On-premise or your cloud. Your hardware, your rules.

2. INTEGRATE

Wire into your existing systems and workflow.

3. TRAIN

Onboard your team end-to-end — ops, review, troubleshoot.

4. HANDOFF

Documented, supported, owned by you.

Typical handoff: 4-8 weeks from contract to first end-to-end run on your data.



Next steps

Pilot dataset

Size, content type, sample source

Integration scope

Systems and pipelines to wire into

Timeline

Weeks to first end-to-end run on your data

Success criteria

What good looks like for your team

Customize per-client during prep — placeholder copy intentionally generic.



Thank You

Questions & Discussion

1,497 segments • 6 quality signals • 5 failure categories • 13 experiments
8-stage pipeline • 37 bugs fixed • 8 research reports

